CHAPTER 9

STRUCTURED QUERY LANGUAGE (SQL)

CHAPTER ESSENCE AT A GLANCE:
Master Structured Query Language (SQL) for relational schema definition (DDL), record manipulation (DML), integrity constraints, aggregate queries, and joins.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Declarative Sublanguage Divisions:** SQL separates database operations into distinct functional divisions: data definition language for creating and altering schema structures, and data manipulation language for querying and modifying stored records.

• **Declarative Retrieval and Filtering:** Relational data queries select specific columns from designated tables, applying conditional evaluation predicates to eliminate non-matching record rows before delivering the result set.

• **Aggregations and Group Partitions:** Standard aggregate functions calculate summary statistics across multiple record rows, while grouping clauses partition data into distinct subsets for collective evaluation.

• **Row Filtering versus Group Filtering:** The standard filtering clause evaluates and filters individual record rows prior to aggregation, whereas the post-aggregation filtering clause operates strictly on assembled group summaries.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

DDL: CREATE, ALTER, DROP | DML: SELECT, INSERT, UPDATE, DELETE

SQL sublanguage classification: structure versus data

FROM → WHERE → GROUP BY → HAVING → SELECT → ORDER BY

Logical processing and execution sequence of an SQL query

WHERE: Filters Rows Prior to Grouping ⟷ HAVING: Filters Groups After Aggregation

Crucial distinction between row and group filtering clauses

$$\text{Average Column Value} = \frac{\text{SUM(column)}}{\text{COUNT(column)}}$$

SQL AVG() aggregate mathematical derivation

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** DDL commands (CREATE, ALTER, DROP) modify relational structure; DML commands (SELECT, INSERT, UPDATE, DELETE) modify relation data.

• **Alert:** WHERE filters individual rows before grouping; HAVING filters aggregated groups after applying GROUP BY.

• **Alert:** Strings and dates in SQL queries must always be enclosed in single quotes ('...').

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Classify SQL commands into DDL, DML, DCL, and TCL with two examples for each category. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Classification):** DDL (Data Definition Language): CREATE, ALTER, DROP, TRUNCATE. DML (Data Manipulation Language): SELECT, INSERT, UPDATE, DELETE. DCL (Data Control Language): GRANT, REVOKE. TCL (Transaction Control Language): COMMIT, ROLLBACK, SAVEPOINT. **[3 Marks]**

• **Key Exam Result:** DDL (schema), DML (data records), DCL (permissions), TCL (transactions)

DDL: CREATE, ALTER, DROP | DML: SELECT, INSERT, UPDATE, DELETE

SQL sublanguage classification: structure versus data

2. Explain common data types in SQL: INT, CHAR, VARCHAR, DATE, and DECIMAL. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Data Types):** INT or INTEGER: Whole numbers without decimals. CHAR(n): Fixed-length character string (padded with spaces). VARCHAR(n): Variable-length character string (memory efficient). DATE: Date in 'YYYY-MM-DD' format. DECIMAL(p, s): Exact fixed-point numbers with p total digits and s decimal places. **[2 Marks]**

• **Key Exam Result:** INT, CHAR, VARCHAR, DATE, DECIMAL

DDL: CREATE, ALTER, DROP | DML: SELECT, INSERT, UPDATE, DELETE

SQL sublanguage classification: structure versus data

3. Write SQL syntax to create a table 'STUDENT' with primary key and check constraints. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (SQL CREATE TABLE):** CREATE TABLE STUDENT (n RollNo INT PRIMARY KEY,n Name VARCHAR(30) NOT NULL,n Gender CHAR(1),n Dob DATE,n TotalMarks DECIMAL(5,2)n); **[3 Marks]**

• **Key Exam Result:** CREATE TABLE STUDENT (RollNo INT PRIMARY KEY, Name VARCHAR(30) NOT NULL, ...);

DDL: CREATE, ALTER, DROP | DML: SELECT, INSERT, UPDATE, DELETE

SQL sublanguage classification: structure versus data

4. Explain table constraints in SQL: PRIMARY KEY, NOT NULL, UNIQUE, CHECK, and DEFAULT. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Constraints):** PRIMARY KEY: Uniquely identifies records, rejects duplicates and NULLs. NOT NULL: Ensures column cannot hold a NULL value. UNIQUE: Rejects duplicate values (allows one NULL). CHECK: Enforces condition on inserted values (CHECK (TotalMarks >= 0)). DEFAULT: Assigns default value if none provided (DEFAULT 'Kerala'). FOREIGN KEY REFERENCES Parent(Key): Enforces referential integrity. **[3 Marks]**

• **Key Exam Result:** PRIMARY KEY, NOT NULL, UNIQUE, CHECK, DEFAULT, FOREIGN KEY

DDL: CREATE, ALTER, DROP | DML: SELECT, INSERT, UPDATE, DELETE

SQL sublanguage classification: structure versus data

5. Explain SQL DML commands: INSERT, UPDATE, and DELETE with syntax and examples. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Commands):** INSERT INTO STUDENT VALUES (101, 'Adithya', 'M', '2007-05-14', 485.50); UPDATE STUDENT SET TotalMarks = TotalMarks + 10 WHERE RollNo = 101; DELETE FROM STUDENT WHERE RollNo = 101; **[4 Marks]**

• **Key Exam Result:** INSERT INTO ... VALUES, UPDATE ... SET, DELETE FROM ... WHERE

$$\text{DDL: CREATE, ALTER, DROP} \quad | \quad \text{DML: SELECT, INSERT, UPDATE, DELETE}$$

SQL sublanguage classification: structure versus data

6. Explain clauses used with SELECT statement: WHERE, DISTINCT, ORDER BY (ASC/DESC). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (SELECT Clauses):** SELECT DISTINCT Dept FROM EMPLOYEE WHERE Salary > 50000 ORDER BY Name ASC; WHERE filters rows; DISTINCT removes duplicates; ORDER BY sorts in ASC/DESC order. **[4 Marks]**

• **Key Exam Result:** SELECT DISTINCT ... FROM ... WHERE ... ORDER BY ...

$$\text{FROM} \rightarrow \text{WHERE} \rightarrow \text{GROUP BY} \rightarrow \text{HAVING} \rightarrow \text{SELECT} \rightarrow \text{ORDER BY}$$

Logical processing and execution sequence of an SQL query

7. Differentiate between CHAR and VARCHAR data types in SQL. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Difference):** CHAR(n): Fixed length. If string is shorter than n, it is padded with spaces up to n bytes. Wastes memory. Faster execution. VARCHAR(n): Variable length. Stores only actual characters plus 1-2 length bytes. Saves disk space. Slightly slower. **[2 Marks]**

• **Key Exam Result:** CHAR (fixed length, padded) vs VARCHAR (variable length, compact)

$$\text{DDL: CREATE, ALTER, DROP} \quad | \quad \text{DML: SELECT, INSERT, UPDATE, DELETE}$$

SQL sublanguage classification: structure versus data